

THE BLUE PLANET

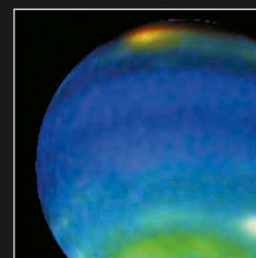
This image of Neptune, which was taken by Voyager 2 on August 19, 1989, reveals the planet's dynamic atmosphere. The Great Dark Spot, which is almost as big as Earth, lies in the center of the planet's disk. A little dark spot and, just above it, the fast-moving cloud feature named the Scooter, are visible on the west limb. A band of cloud stretches across the northern polar region.

NEPTUNE PROFILE

	
AVERAGE DISTANCE FROM THE SUN 2.8 billion miles (4.5 billion km)	ROTATION PERIOD 16.11 hours
CLOUDTOP TEMPERATURE −360°F (−218°C)	ORBITAL PERIOD (LENGTH OF YEAR) 164.8 years
DIAMETER 30,760 miles (49,532 km)	MASS (EARTH = 1) 17.1
VOLUME (EARTH = 1) 57.74	GRAVITY AT CLOUDTOPS (EARTH = 1) 1.13
NUMBER OF MOONS 14	SIZE COMPARISON
OBSERVATION Even at its maximum magnitude of 7.8, Neptune is beyond naked-eye visibility. Binoculars or a small telescope will show it as a starlike point of light. Its long orbit means it takes years to move through each zodiacal constellation.	EARTH  NEPTUNE 

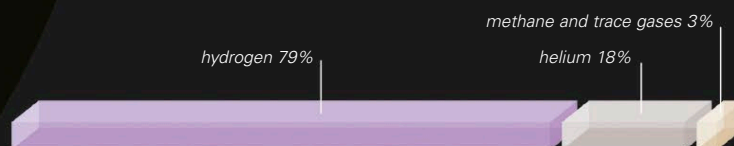
ATMOSPHERE AND WEATHER

Neptune is a perplexing place. For a planet so far from the Sun, it has a surprisingly dynamic atmosphere that exhibits colossal storms and super-fast winds. The heat Neptune receives from the Sun is not enough to drive its weather. The atmosphere may be warmed from below by Neptune's internal heat source, and this is the trigger for larger-scale atmospheric changes. The white bands that encircle the planet are cloud cover, produced when the heated atmosphere rises and then condenses, forming clouds. The winds are most ferocious in the equatorial regions, where they blow westward and reach a staggering 1,340 mph (2,160 kph). Gigantic, dark, stormlike features accompanied by bright, high-altitude clouds appear and then disappear. One, the Great Dark Spot, was seen by Voyager 2 in 1989. When the Hubble Space Telescope looked for the storm in 1996, it had disappeared.



CLOUDS OVER NEPTUNE

Neptune's atmosphere lies in bands, which are parallel to the equator. The bright patches are high-altitude clouds, floating above the blue methane layer.

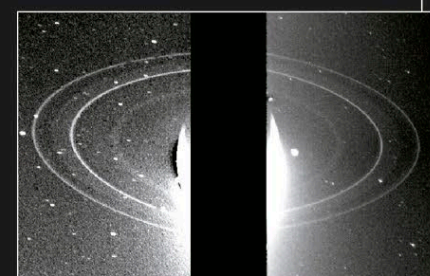


COMPOSITION OF ATMOSPHERE

Neptune's atmosphere is made mostly of hydrogen. But it is the methane that gives the planet its deep blue color, absorbing red light and reflecting blue.

RINGS AND MOONS

The first indication that Neptune has a ring system came in the 1980s, when stars were seen to blink on and off near the planet's disk. Intriguingly, Neptune seemed to have ring arcs. The mystery was solved when Voyager 2 discovered that Neptune has a ring system with an outer ring so thinly populated that it does not dim starlight but contains three dense regions that do. Neptune has five sparse yet complete rings; moving in from the outer Adams ring, they are Arago, Lassell, Le Verrier, and Galle. A sixth, unnamed partial ring lies within Adams. The rings are made of tiny pieces of unknown composition, which together would make a body just a few miles across. The material is believed to have come from nearby moons. Five of Neptune's 14 moons are within the ring system. It is one of the moons, Galatea, that prevents the arc material from spreading uniformly around the Adams ring. Only one of the 14, Triton, is of notable size. Triton and Nereid were discovered before the days of space probes. Six small moons have been discovered since 2002, and more will probably be found.



THE RINGS OF NEPTUNE

Two Voyager 2 images placed together reveal Neptune's ring system. The two bright rings are Adams and Le Verrier. The faint Galle ring is innermost, and the diffuse band, Lassell, is visible between the two bright ones.

NEPTUNE'S MOONS

